Aarush Ram Anandh

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EDUCATION

Ashoka University, Sonipat **2024–2028**
B.Sc. (Hons.) Mathematics and Computer Science

Relevant coursework: Real Analysis, Linear Algebra, Probability and Statistics, Statistical Inference, Machine Learning, Algorithms, Discrete Mathematics; currently pursuing Algebra I, Complex Analysis, Artificial Intelligence, Reinforcement Learning and Symbolic Logic.

RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved colour fidelity in latent diffusion models, achieving $\Delta E_{00} < 1$ and approximately **4× better colour accuracy** than evaluated open-source baselines; this work is reported in the preprint listed under Publication.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, a controllable optimal-transport framework with control-conditioned probes, projections and slice aggregation. Paper submitted to JMLR.
- Studying metricity, admissibility, stability, minimality and optimal probe/readout construction; implementing computational methods in **C++** and machine-verified specifications and proofs in **Lean**.
- Built an image-processing pipeline that generated **20,000+** training inputs for lab use. [GitHub]

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Developed patient-day preprocessing for **191 patients and 15,691 patient-days**, processing **12.1 million heart-rate and 736,000 HRV readings** alongside sleep, temperature and activity data.
- Implemented HR/HRV smoothing, residual extraction, change-point features and windowed aggregation for **XGBoost, CatBoost, RNN and LSTM** classifiers, with 1–7-day input-window ablations.
- Implemented **patient-wise 80:20 train/test evaluation across five seeds** and patient-bootstrap confidence intervals, keeping same-day detection separate from future-onset prediction.
- Built experimental routes for frozen **Timestral, TimesFM and Chronos-2** forecast-surprise features and **MOMENT/MantisV2** embeddings; these routes were implemented but not yet run in the reported cohort evaluation.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling **2026**
Colour-preserving methods for latent diffusion; arXiv:2601.17259.

ADDITIONAL TECHNICAL & TEACHING EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Raised rare-class correct detection from **40% to 80%** (**+40 percentage points**) in an **RT-DETR** detection, classification and tracking pipeline for **18 waste-material classes**.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning to improve rare-class performance; evaluated detector failure modes with crop-based refinement.
- Helped set up a **near-infrared (NIR) waste-segregation system** for visually similar plastics, handling parts procurement and running preliminary experiments.

Lodha Genius Programme, Ashoka University **Jun–Jul 2025**
Teaching Assistant

- Supported four faculty members in an IUCAA telescope workshop and an IoT workshop for **200+ students**.

Yulu Bikes **Apr–May 2023**
Research / Engineering Project

- Modelled electric-vehicle dynamics in **MATLAB/Simulink**, testing the effect of vehicle parameters and operating conditions on system performance.

PROJECTS

MEIKURAL: Hindustani Raag Recognition **Ongoing**

Audio Classification / Signal Processing [GitHub]

- Developing a **Python/PyTorch** pipeline to classify **seven Hindustani raags** from audio, using recordings from the Saraga Hindustani dataset.
- Implemented resampling, RMS normalisation, overlapping **30-second windows**, STFT feature extraction and CNN classification; explored per-window tonic alignment, full-recording tonic alignment and native-frequency spectrogram representations.
- Trained a native-STFT CNN on **3,471 audio windows** over 30 CUDA epochs; built interactive clip inference, prediction timelines and recording-level and window-level confusion-matrix reports.

Monthly Equity Return Prediction **Completed May 2026**

Introduction to Machine Learning Coursework / Team Project [GitHub]

- Collaborated with **Tanush Seal** on a company-month financial forecasting pipeline combining a **CatBoost** baseline with residual-correction models.
- Engineered lagged features, rolling statistics, momentum and volatility measures, categorical features and missingness indicators for financial panel data.
- Compared **seven residual approaches**: Gaussian HMM, Transformer, MLP, CNN, Ridge, Elastic Net and CatBoost; developed chronological evaluation splits, company-level error diagnostics and prediction dashboards.

Reinforcement Learning for Pokemon Battles **Ongoing**

Reinforcement Learning Coursework / Two-Person Team [GitHub]

- Collaborating on a controlled study of move-outcome variability and agent learning in **4v4 Pokemon Showdown battles**, comparing low- and high-variance move sets.
- Co-developed a shared **Gymnasium/poke-env** environment with **763 observation features**, legality masks over 26 actions and potential-based reward shaping, with **MaskablePPO** and masked **DQN** agents.
- Evaluated PPO across **two team variants and five seeds per variant**, with 200,000 decision steps per run; recorded final win rates of **99.7–100%** over 1,020 battles per run against the fixed heuristic opponent across three opponent teams.
- Analysed learning curves, convergence, Wilson confidence intervals and exact permutation tests; found no statistically significant learning-speed difference between team variants in the tested setup.

CDR-Decomposition Pipeline **Jan 2026**

Vector Graphics / Training-Data Preparation [GitHub]

- Built a **Python** pipeline to extract structural die-line vectors from flexible-packaging **CorelDRAW (CDR)** artwork, supporting generation of **20,000+ training inputs** for internal lab use.
- Automated headless **LibreOffice** conversion from CDR to SVG, parsed SVG/XML to separate embedded raster images, and applied luminance-based greyscale conversion and colour inversion.
- Isolated high-contrast structural elements through black–white correspondence checks and geometric filtering of lines and rectangles; retained intermediate SVG outputs and extracted-raster metadata for inspection.

Shooting Stability Trainer **Sep 2025**

Computer Vision / Interactive Training Prototype [GitHub]

- Prototyped a webcam-based, low-cost alternative to SCATT-style motion tracing for rifle and pistol training using **Python, OpenCV, Flask-SocketIO and JavaScript**.
- Tracked a coloured marker through HSV thresholding, contour filtering and centroid estimation; streamed marker coordinates and segmentation masks to a browser interface.
- Implemented adjustable HSV thresholds, two-point scale calibration, aim-centre selection and manual motion-trace recording on a virtual target.

Hyperspectral Waste Analysis **Sep 2026–Present**

Ongoing CGSW Experiment

- Conducting a hyperspectral waste-analysis experiment within the **Controlled Generalised Sliced Wasserstein (CGSW)** research described under Research Experience, using image cubes with **224 spectral bands** and segmentation masks across **seven waste categories**.
- Constructing object-level spectral measures to evaluate **CGSW** for class separability, alongside sliced optimal-transport distances.

TECHNICAL SKILLS

Programming: Python, C++, C, OCaml, Bash **Machine learning:** PyTorch, Diffusers, TensorFlow

Formal methods: Lean **Research areas:** Optimal transport, diffusion models, computer vision, time-series analysis

Scientific computing: MATLAB/Simulink, OpenFOAM